Red Cell Disorders

Lec. 14

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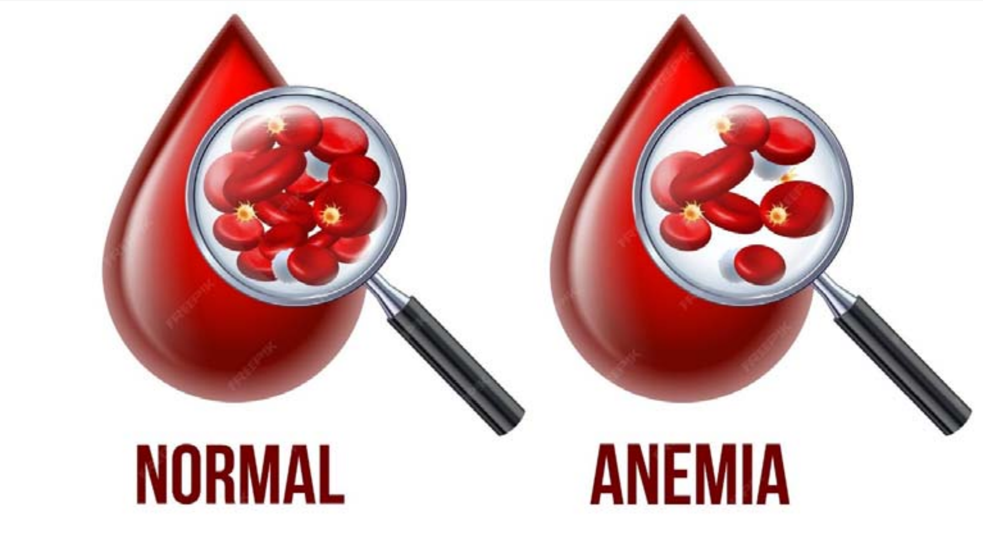
Red Cell Disorders

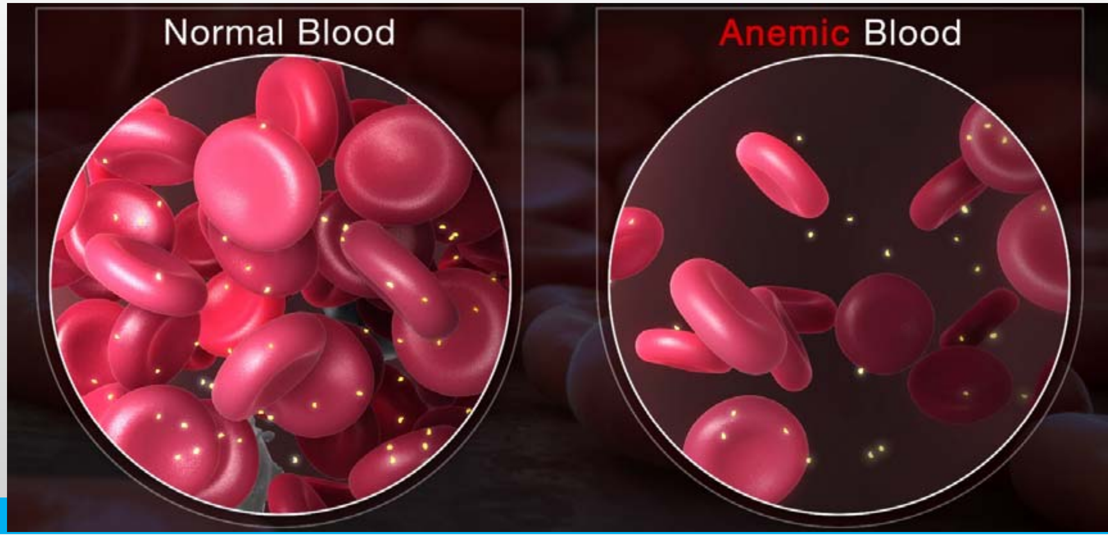
- Disorders of red cells can result in anemia or, less commonly, polycythemia (an increase in red cells also known as erythrocytosis).
- **Anemia** is defined as a reduction in the oxygen-transporting capacity of blood, resulting from a decrease in the red cell mass to subnormal levels.



Classification of anemia

- Anemia can stem from bleeding, increased red cell destruction, or decreased red cell production. These mechanisms serve as one basis for classifying anemia into :

1) Anemia of blood loss
2) Hemolytic anemias
3) Anemia of diminished erythropoiesis



Anemia of blood loss

- Anemia of blood loss can be divided into anemia caused by acute bleeding (hemorrhage) and anemia caused by chronic blood loss .

Acute bleeding:

The effects of acute bleeding are mainly due to the loss of intravascular volume, which if massive can lead to cardiovascular collapse, shock, and death. If blood loss exceeds 20% of blood volume, the immediate threat is hypovolemic shock rather than anemia.

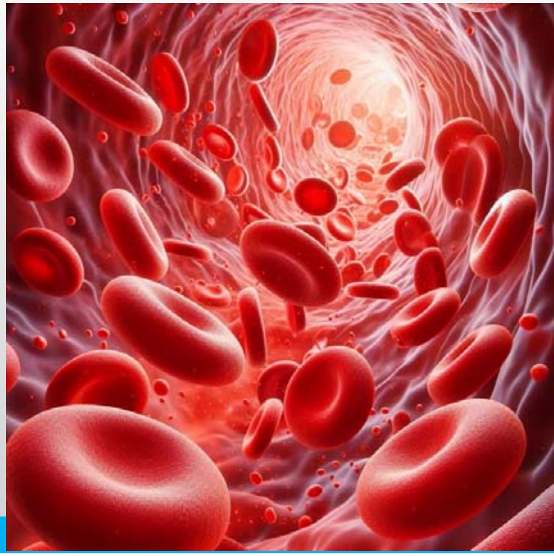
Chronic blood loss:

- With chronic blood loss, iron stores are gradually depleted. The main causes of chronic haemorrhage are:

1-Diseases of gastrointestinal tract particularly peptic ulceration , carcinoma of the stomach and carcinoma of the colon

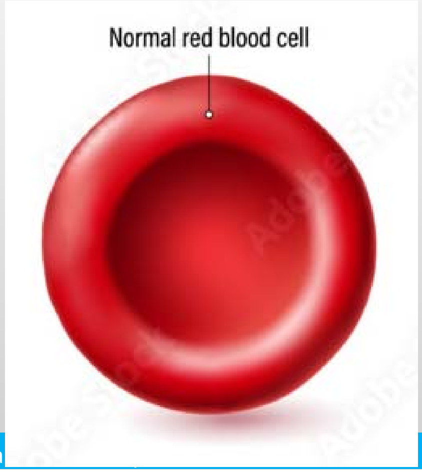
2-Menorrhagia

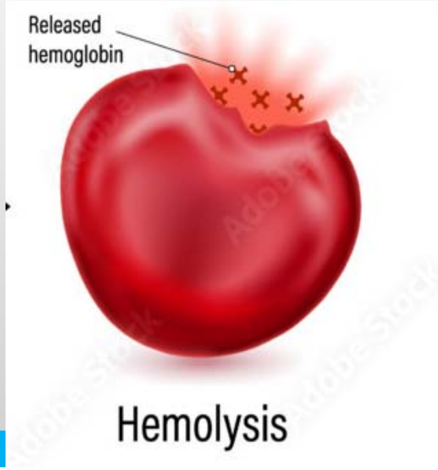
3-Lesions in the urinary tract



Hemolytic anemia

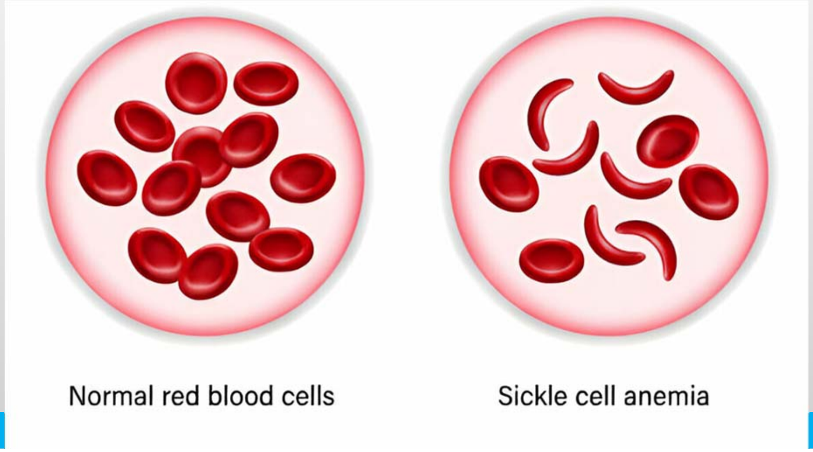
- Hemolytic anemias are a diverse group of disorders that have as a common feature accelerated red cell destruction (hemolysis).
- Examples of hemolytic anemias are : sickle cell anemia and thalassemia .





Sickle cell anemia:

- Sickle cell anemia is the most common familial hemolytic anemia , is caused by a mutation in β-globin that creates sickle hemoglobin (HbS). Sickle cell anemia is caused by a single amino acid substitution in β-globin that results in a tendency for deoxygenated HbS to self-associate into polymers. These polymers distort the red cell, which assumes an elongated crescentic, or sickle, shape .Normal hemoglobins are tetramers composed of two pairs of similar chains.



Anemia of diminished erythropoiesis

This category includes :

1-Anemias that are caused by an inadequate dietary supply of substances that are needed for hematopoiesis : iron ,folic acid, vitamin B12.

Example : Iron deficiency anaemia

2- Other disorders that suppress erythropoiesis include those associated with bone marrow failure ((aplastic anemia)

3- Myelophthisic anemia :the replacement of the bone marrow by tumor or inflammatory cells .

Iron deficiency anaemia:

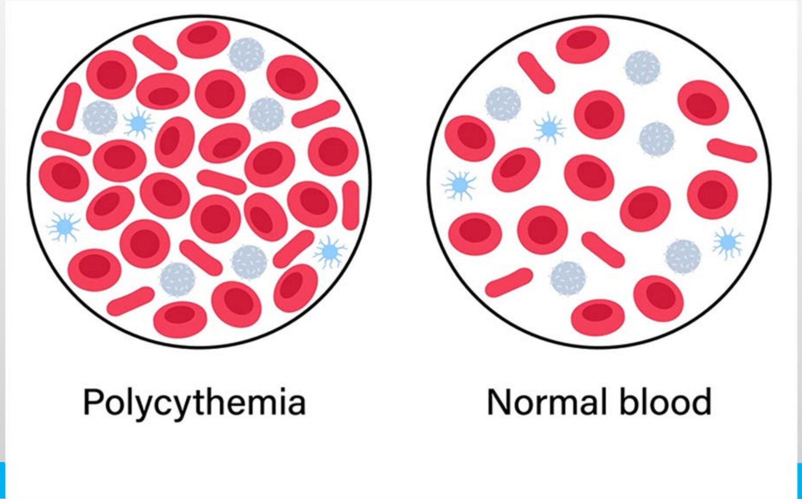
- Iron deficiency anaemia is the most common cause of anaemia, iron is abundant in meat, vegetables, eggs and dairy foods. Deficiency of iron is the most common nutritional deficiency in the world and results in clinical signs and symptoms that are mostly related to anemia . Deficiency of iron results in insufficient hemoglobin synthesis.

- Chronic blood loss is also important cause of

  iron deficiency anemia .

Polycythemia

- Polycythemia, or erythrocytosis, denotes an increase in red cells per unit volume of peripheral blood. It may be absolute (defined as an increase in total red cell mass) or relative. Relative polycythemia results from dehydration, such as occurs with water deprivation, prolonged vomiting, diarrhea, or the excessive use of diuretics.

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